ANS 1.

Point	Distance from Centroid 1 (2,6)	Distance from Centroid 2 (5,10)	Distance from Centroid 3 (6,11)	Assigned Cluster
A1 (2,10)	4	3	4.123106	Cluster 2
A2 (2,6)	0	5	6.403124	Cluster 1
A3 (11,11)	10.29563	6.082763	5	Cluster 3
A4 (6,9)	5	1.414214	2	Cluster 2
A5 (6,4)	4.472136	6.082763	7	Cluster 1
A6 (1,2)	4.123106	8.944272	10.29563	Cluster 1
A7 (5,10)	5	0	1.414214	Cluster 2
A8 (4,9)	3.605551	1.414214	2.828427	Cluster 2
A9 (10,12)	10	5.385165	4.123106	Cluster 3
A10 (7,5)	5.09902	5.385165	6.082763	Cluster 1
A11 (9,11)	8.602325	4.123106	3	Cluster 3
A12 (4,6)	2	4.123106	5.385165	Cluster 1
A13 (3,10)	4.123106	2	3.162278	Cluster 2
A14 (3,8)	2.236068	2.828427	4.242641	Cluster 1
A15 (6,11)	6.403124	1.414214	0	Cluster 3

- In cluster 1, we have 6 points i.e. A2 (2,6), A5 (6,4), A6 (1,2), A10 (7,5), A12 (4,6), A14 (3,8). To calculate the new centroid for cluster 1, we will find the mean of the x and y coordinates of each point in the cluster. Hence, the new centroid for cluster 1 is (3.833, 5.167).
- In cluster 2, we have 5 points i.e. A1 (2,10), A4 (6,9), A7 (5,10) , A8 (4,9), and A13 (3,10). Hence, the new centroid for cluster 2 is (4, 9.6)
- In cluster 3, we have 4 points i.e. A3 (11,11), A9 (10,12), A11 (9,11), and A15 (6,11). Hence, the new centroid for cluster 3 is (9, 11.25).

ANS 2.

Calculating distance between data point A and other data points as:

$d(A, B) = \sqrt{(4-3)^2 + (6-7)^2} = \sqrt{(1)^2 + (-1)^2} = \sqrt{1+1} = \sqrt{2} = 1.4142$

$d(A, C) = \sqrt{(5-3)^2 + (5-7)^2} = \sqrt{(2)^2 + (-2)^2} = \sqrt{4+4} = \sqrt{8} = 2.8284$

$d(A, D) = \sqrt{(6-3)^2 + (4-7)^2} = \sqrt{(3)^2 + (-3)^2} = \sqrt{9+9} = \sqrt{18} = 4.2426$

$d(A, E) = \sqrt{(7-3)^2 + (3-7)^2} = \sqrt{(4)^2 + (-4)^2} = \sqrt{16+16} = \sqrt{32} = 5.6569$

$d(A, F) = \sqrt{(6-3)^2 + (2-7)^2} = \sqrt{(3)^2 + (-5)^2} = \sqrt{9+25} = \sqrt{34} = 5.8310$

$d(A, G) = \sqrt{(7-3)^2 + (2-7)^2} = \sqrt{(4)^2 + (-5)^2} = \sqrt{16+25} = \sqrt{41} = 6.4031$

$d(A, H) = \sqrt{(8-3)^2 + (4-7)^2} = \sqrt{(5)^2 + (-3)^2} = \sqrt{25+9} = \sqrt{34} = 5.8310$

Calculating distance between data point B and other data points as:

$d(B, C) = \sqrt{(5-4)^2 + (5-6)^2} = \sqrt{(1)^2 + (-1)^2} = \sqrt{1+1} = \sqrt{2} = 1.4142$

$d(B, D) = \sqrt{(6-4)^2 + (4-6)^2} = \sqrt{(2)^2 + (-2)^2} = \sqrt{4+4} = \sqrt{8} = 2.8284$

$d(B, E) = \sqrt{(7-4)^2 + (3-6)^2} = \sqrt{(3)^2 + (-3)^2} = \sqrt{9+9} = \sqrt{18} = 4.2426$

$d(B, F) = \sqrt{(6-4)^2 + (2-6)^2} = \sqrt{(2)^2 + (-4)^2} = \sqrt{4+16} = \sqrt{20} = 4.4721$

$d(B, G) = \sqrt{(7-4)^2 + (2-6)^2} = \sqrt{(3)^2 + (-4)^2} = \sqrt{9+16} = \sqrt{25} = 5$

$d(B, H) = \sqrt{(8-4)^2 + (4-6)^2} = \sqrt{(4)^2 + (-2)^2} = \sqrt{16+4} = \sqrt{20} = 4.4721$

Calculating distance between data point C and other data points as:

$d(C, D) = \sqrt{(6-5)^2 + (4-5)^2} = \sqrt{(1)^2 + (-1)^2} = \sqrt{1+1} = \sqrt{2} = 1.4142$

$d(C, E) = \sqrt{(7-5)^2 + (3-5)^2} = \sqrt{(2)^2 + (-2)^2} = \sqrt{4+4} = \sqrt{8} = 2.8284$

$d(C, F) = \sqrt{(6-5)^2 + (2-5)^2} = \sqrt{(1)^2 + (-3)^2} = \sqrt{1+9} = \sqrt{10} = 3.1623$

$d(C, G) = \sqrt{(7-5)^2 + (2-5)^2} = \sqrt{(2)^2 + (-3)^2} = \sqrt{4+9} = \sqrt{13} = 3.6056$

$d(C, H) = \sqrt{(8-5)^2 + (4-5)^2} = \sqrt{(3)^2 + (-1)^2} = \sqrt{9+1} = \sqrt{10} = 3.1623$

Calculating distance between data point D and other data points as:

$d(D, E) = \sqrt{(7-6)^2 + (3-4)^2} = \sqrt{(1)^2 + (-1)^2} = \sqrt{1+1} = \sqrt{2} = 1.4142$

$d(D, F) = \sqrt{(6-6)^2 + (2-4)^2} = \sqrt{(0)^2 + (-2)^2} = \sqrt{0+4} = \sqrt{4} = 2$

$d(D, G) = \sqrt{(7-6)^2 + (2-4)^2} = \sqrt{(1)^2 + (-2)^2} = \sqrt{1+4} = \sqrt{5} = 2.2361$

$d(D, H) = \sqrt{(8-6)^2 + (4-4)^2} = \sqrt{(2)^2 + (0)^2} = \sqrt{4+0} = \sqrt{4} = 2$

Calculating distance between data point E and other data points as:

$d(E, F) = \sqrt{(6-7)^2 + (2-3)^2} = \sqrt{(-1)^2 + (-1)^2} = \sqrt{1+1} = \sqrt{2} = 1.4142$

$d(E, G) = \sqrt{(7-7)^2 + (2-3)^2} = \sqrt{(0)^2 + (-1)^2} = \sqrt{0+1} = \sqrt{1} = 1$

$d(E, H) = \sqrt{(8-7)^2 + (4-3)^2} = \sqrt{(1)^2 + (1)^2} = \sqrt{1+1} = \sqrt{2} = 1.4142$

Calculating distance between data point F and other data points as:

$d(F, G) = \sqrt{(7-6)^2 + (2-2)^2} = \sqrt{(1)^2 + (0)^2} = \sqrt{1+0} = \sqrt{1} = 1$

$d(F, H) = \sqrt{(8-6)^2 + (4-2)^2} = \sqrt{(2)^2 + (2)^2} = \sqrt{4+4} = \sqrt{8} = 2.8284$

Calculating distance between data point G and other data points as:

$d(G, H) = \sqrt{(8-7)^2 + (4-2)^2} = \sqrt{(1)^2 + (2)^2} = \sqrt{1+4} = \sqrt{5} = 2.2361$

Data Points	A	B	C	D	E	F	G	H
A	0	1.4142	2.8284	4.2426	5.6569	5.831	6.4031	5.831
B		0	1.4142	2.8284	4.2426	4.4721	5	4.4721
C			0	1.4142	2.8284	3.1623	3.6056	3.1623
D				0	1.4142	2	2.2361	2
E					0	1.4142	1	1.4142
F						0	1	2.8284
G							0	2.2361
H								0

N(A) = {B}; — — — — — — -→ because distance of B is <= 2.5 with A
N(B) = {A, C}; — — — — — → because distance of A and C is <= 2.5 with B
N(C) = {B, D}; — — — — —→ because distance of B and D is <=2.5 with C
N(D) = {C, E, F, G, H}; — → because distance of C, E, F,G and H is <=2.5 with D
N(E) = {D, F, G, H}; — — → because distance of D, F, G and H is <=2.5 with E
N(F) = {D, E, G}; — — — — → because distance of D, E and G is <=2.5 with F
N(G) = {D, E, F, H}; — — -→ because distance of D, E, F and H is <=2.5 with G
N(H) = {D, E, G}; — — — — → because distance of D, E and G is <=2.5 with H

Here, data points A, B and C have neighbors <= MinPts (i.e. 3) so can't be considered as core points. Since they belong to the neighborhood of other data points, hence there exist no outliers in the given set of data points.

Data points D, E, F, G and H have neighbors >= MinPts (i.e. 3) and hence are the core data points.

ANS 3. With first two objects as initial centroids:
Centroid for first cluster c1 = (185, 72)
Centroid for second cluster c2 = (170, 56)

Instance	X	Y	Distance(C1)	Distance(C2)	Cluster
1	185	72			c1
2	170	56			c2
3	168	60	√433	√20	c2
4	179	68	√52	√225	c1
5	182	72	√9	√400	c1
6	188	77	√34	√765	c1

Distance of each data points from cluster centroids

Answer 4

(a) To find the total cost, we calculate the Manhattan distance from the medoid B(3,7) to every other point and sum them up.

- **Distance(B to A):** $|3-2|+|7-6|=1+1=2$
- **Distance(B to C):** $|3-4|+|7-6|=1+1=2$
- **Distance(B to D):** $|3-3|+|7-5|=0+2=2$
- **Distance(B to E):** $|3-8|+|7-2|=5+5=10$

Total Initial Cost = $2+2+2+10=16$

(b) calculating the total cost from the potential new medoid C(4,6) to all other points.

- **Distance(C to A):** $|4-2|+|6-6|=2+0=2$
- **Distance(C to B):** $|4-3|+|6-7|=1+1=2$
- **Distance(C to D):** $|4-3|+|6-5|=1+1=2$
- **Distance(C to E):** $|4-8|+|6-2|=4+4=8$

New Total Cost = $2+2+2+8=14$

(c) Yes, the swap should be made.

The new total cost with C as the medoid (14) is lower than the initial total cost with B as the medoid (16). Since the algorithm's goal is to minimize the total intra-cluster distance, it will accept this change.

Answer 5

(a) The algorithm would find **2 clusters**.

The `epsilon` would be large enough to connect the 10 points within Cluster A and the 10 points within Cluster B, but small enough so that the two clusters do not connect to each other. Since `MinPts = 3` and each cluster is dense, core points would be easily identified, and the clusters would form. The isolated points, **X(1,1)** and **Y(12,10)**, would not have enough neighbors to form a cluster and would be classified as **noise**.

(b) The algorithm would likely find only **1 cluster** containing all the data points.

A very large `epsilon` would create a large neighborhood around each point. The radius would be large enough to connect Cluster A, Cluster B, and the outliers (X and Y) together. As a result, all points would be considered

density-reachable from each other and would be merged into a single large cluster.

(c) The algorithm would find **0 clusters**, and all points would be classified as noise.

The condition to be a core point is that a point must have at least `MinPts` neighbors within its `epsilon` radius. Since the largest cluster only contains 10 points, no point in the dataset can meet the requirement of having 12 neighbors. Without any core points, no clusters can be formed, and every point in the dataset would be considered an outlier.